



A novel weighted anisotropic total variational model for image applications

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Abstract

In this paper, a novel weighted anisotropic total variational (WATV) model is proposed for image restoration. In this model, a weight function is defined to process degraded images, which is introduced into the ℓ_1 norm-based regularization. In order to efficiently compute the restored images, the alternating direction method of multipliers (ADMM) is explored and the according convergence is analyzed briefly. The proposed model is applied to recover images with blurring and noise. Compared experiments of deblurring, denoising as well as restoration are conducted to show the effectiveness and efficiency of the proposed WATV model.

Keywords Image restoration · Total variational · Weight function · ADMM

1 Introduction

Image restoration is an extensively studied problem in field of image processing. The goal of image restoration is recovering clear, sharp images from degraded versions and meanwhile keeping more image details, such as textures and edges. So it is usually studied together with image denoising and deblurring. It is easy to find that some details of an image will be damaged when it is polluted by blurring or noise. Then some important information of images get disappeared even the object inside the image is difficult to recognize. Therefore, in the process of restoration, it requires both of structure and detail preservation, which respectively corresponds to fidelity and regularization in the mathematical model. In this sense, the restoration problem is often modeled by

$$\min_u F(u, f) + \lambda R(u) \quad (1)$$

where f is the degraded image, u is restored from f , $F(u, f)$ is a fidelity term to keep the image structures, $R(u)$ is a regularization term, λ is a balanced parameter. Regularization is essential to image processing and detail preservation. What is more, the lack of prior information that makes the restoration problem ill-posed also needs good regularization. So many studies focus on the regularization for image restoration and other image processing problems, such as denoising [1], deblurring [2], decomposition [3,4], segmentation [5,6], reconstruction [7,8], etc.

During the regularization, the total variational is widely explored. In [9], a total variational (TV) model or Rudin–Osher–Fatemi (ROF) model is proposed to image denoising. The TV-based models are very efficient in calculation. They also perform well in edge preservation. The main disadvantage of TV-based models is staircase effect [10,11]. Many modified models are proposed to deal with it. In [12,13], higher order extensions of the total variational are proposed. However, they may result in blur edges. Authors of [14,15] consider fractional order derivative and incorporate it into their models to explore image features in sense of fractional differentiation. There are some other models work for this, such as total generalized variational (TGV) [16], hybrid TV [17], nonlocal TV [18], etc. These models above mainly consider different orders of differentiation to explore the denoising problem. These models are effective to reduce the staircase effect. Different from the original TV model which is isotropic, during the process anisotropy [19–21] is also

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considered. The isotropic processes images in the same way in x and y directions of gradient, while the anisotropic considers them separately and gets better performance.

Besides, the sparsity of image gradient is considered as the regularization in restoration models, such as ℓ_0 norm-based [22,23], ℓ_1 norm-based [24], ℓ_p norm-based [25,26]. Some authors also explore the weights on regularization. In [27], $\ell_1 + \ell_2^2$ -based regularization is explored. In [28], they employ the regularization in form of $L_1 - \alpha L_2$. The weight is conducted on the combination of ℓ_1 and ℓ_2 norms. They work well in image restoration; however, they may take more computational time due to the more complex regularizations in their models. In addition, newly developed learning-based methods are also explored, which introduced different priors to modelling, such as dark channel prior [29,30], hybrid deep prior [31], tri-segment intensity prior [32], sparse prior [33–35]. In these studies, although training data may be large, networks are trained with different priors and can achieve better results. It can be seen that whether variational or learning-based methods, the prior information is of great importance.

Based on the mentioned studies, detail preservation and efficient calculation are required in restoration problem. The quality of restored images and the computational time depend on properly constructed models and efficient algorithms. Therefore, in order to reduce the staircase of TV-based models and keep image details at the same time, a weight with respected to image gradient is defined and incorporated into a new regularization. Hence, a weighted anisotropic total variational (WATV) model is designed. The weight is formulated as a nonlinear and monotonically increasing function. It assigns different weights to pixels of the image. Larger weights penalize the high shock caused by noise and enhance the blurred edges. Smaller weights help to keep details of smooth areas. The specific definition is shown in Sect. 3.

As for calculation, many methods are explored to solving the regularized models, such as primal dual methods [36], alternating direction method of multipliers (ADMM) [37]. They are popular and widely used in many image problems [38–41].

In this paper, we utilize the ADMM to solve the proposed model. In the frame of ADMM, the augmented Lagrangian function is used to reform the proposed model. The problem is separated into several subproblems and they are calculated respectively.

The paper is organized as follows. In Sect. 2, we describe some notations and give a brief review of some related works; in Sect. 3, the proposed model is formulated, the algorithm ADMM is organized and the convergence is analyzed. In Sect. 4, compared experiments are conducted to demonstrate the effectiveness and efficiency of the proposed model. Finally, we conclude the paper in Sect. 5.

2 Preliminaries

In this section, some notations are declared and some related models for image restoration are summarized.

2.1 Notations

Let $f : \Omega \rightarrow \mathcal{R}$ be a given image that may contain blur and noise, where Ω is the image domain. The given image is often decomposed into the following form

$$f = Bu + n \quad (2)$$

where B is a linear operator, u is the new image that recovered from f , n is the additive noise. Denoting the size of u as $M \times N$, the discrete gradients of u in x and y directions are computed by differences as follows:

$$\begin{aligned} D_x u(i, j) &= \begin{cases} u(i, j) - u(i-1, j), & 1 < i \leq M, 1 \leq j \leq N \\ u(i, j) - u(M, j), & i = 1, 1 \leq j \leq N \end{cases} \\ D_y u(i, j) &= \begin{cases} u(i, j) - u(i, j-1), & 1 \leq i \leq M, 1 < j \leq N \\ u(i, j) - u(i, N), & 1 \leq i \leq M, j = 1 \end{cases} \end{aligned} \quad (3)$$

where (i, j) represents the spacial coordinate of a pixel, D_x and D_y are differential operator in x and y directions respectively. For image gradient Du , the ℓ_1 norm is computed by

$$\|Du\|_1 = \|D_x u\|_1 + \|D_y u\|_1 \quad (4)$$

2.2 L_1 Regularized model

The ℓ_1 norm of image gradient introduced in the well-known TV model or ROF model was proposed by Rudin et al in [9]. It is originally designed for denoising by

$$\min_u \frac{\lambda}{2} \|f - u\|_2^2 + \|Du\|_1. \quad (5)$$

where $\lambda > 0$ is a balanced parameter that controls the trade-off between data fidelity and regularization. In this model, ℓ_1 norm of image gradient is used to regularize images and performs well in maintaining edges during denoising. Here $\|Du\|_1$ is isotropic. It is widely applied in many image problems even though it can generate staircase effect during processing. With the same regularization, for restoration the fidelity of (5) is rewritten as

$$\|Bu - f\|_2^2 \quad (6)$$

where B is a linear operator differently chosen for different requirements. When B takes the identical operator, it

becomes the denoising model. When B is a blur operator, it can process images with blur.

2.3 $L_1 - \alpha L_2$ -based model

In [28], a weighted difference of anisotropic and isotropic total variation is proposed as the regularization

$$J(u) = \|D_x u\|_1 + \|D_y u\|_1 - \alpha \sqrt{|D_x u|^2 + |D_y u|^2} \quad (7)$$

The overall model is constructed by

$$\min_u J(u) + \frac{\lambda}{2} \|Au - f\|_2^2 \quad (8)$$

where A is a linear operator, α and λ are parameters. This regularization helps to increase the sparsity of image gradient and achieve good results in applications of denoising and deblurring. When $\alpha = 0.5$, the L_1 -0.5 L_2 performs better than it takes other values in $[0, 1]$. However, the complexity of the regularization makes it cost more time to compute the model.

2.4 HNSOTV-OGS model

Recently, [42] proposed a hybrid non-convex second-order total variation model with group sparsity (HNSOTV-OGS) for deblurring, in which non-convex higher-order TV and overlapping group sparse regularizers are combined. The HNSOTV -OGS model is formulated by

$$\min_u \frac{\lambda}{2} \|Ku - f\|_2^2 + \phi(\nabla u) + \omega \|\nabla^2 u\|_p^p + \mathcal{I}_c(u) \quad (9)$$

where K is a blurring operator, $\phi(\cdot)$ is the overlapping group sparse, $\mathcal{I}_c$ is a characteristic function. As is discussed by the authors, $p = 0.1$ would be the best value to use. The regularization performs well in deblurring application. However, the complexity of the model takes more the computational time, especially the group size for sparsity regularizer in their algorithm.

3 The proposed model and algorithm

In order to restore images with higher quality and meanwhile achieve efficient calculation, a weighted anisotropic total variational model is proposed for the restoration task.

3.1 Model formulation

The proposed model is formulated as

$$\min_u E(u) = \frac{\lambda}{2} \|Bu - f\|_2^2 + \|w \cdot Du\|_1 \quad (10)$$

where B is a linear operator, $f \in \mathcal{R}^{m \times n \times l}$ is an observed image blurred or simultaneously polluted by noise, λ is a positive parameter to control the balance between two terms in (10), w is the weight defined by a function with respect to image gradient as follows:

$$w(Df) = \frac{c}{1 + \exp^{-|Df|}} \quad (11)$$

where $c > 0$, $|Du|$ is the amplitude of gradient of u . The weight function is nonlinearly monotonic with respect to gradient amplitudes, which adaptively assigns different weights to pixels of an image. So in the basis of anisotropic definition of the ℓ_1 norm, the newly defined weight is adaptive to pixels of the image. Therefore, the weighted anisotropic total variational model is designed. B is blur operator for deblurring and identical operator for denoising.

For convenience, $w(D_x f)$ and $w(D_y f)$ are denoted as w_1 and w_2 , respectively. Two temporary variables u_1 and u_2 are introduced, where $u_1 = w_1 \cdot D_x u$, $u_2 = w_2 \cdot D_y u$. The energy function (10) can be reformulated as the following constrained optimization problem.

$$\begin{aligned} \min_u \quad & \frac{\lambda}{2} \|Bu - f\|_2^2 + \|u_1\|_1 + \|u_2\|_1 \\ \text{s.t.} \quad & u_1 = w_1 \cdot D_x u, \quad u_2 = w_2 \cdot D_y u. \end{aligned} \quad (12)$$

3.2 Algorithm

In order to solve model (12), the widely explored ADMM is utilized. Denoting $w(\nabla_x f)$ and $w(\nabla_y f)$ as w_1 and w_2 , respectively, the augmented Lagrangian of model (12) is given by

$$\begin{aligned} L(u, u_1, u_2; r_1, r_2) = & \frac{\lambda}{2} \|Bu - f\|_2^2 + \|u_1\|_1 \\ & + \|u_2\|_1 + \langle r_1, u_1 - w_1 D_x u \rangle \\ & + \langle r_2, u_2 - w_2 D_y u \rangle + \frac{\tau}{2} \|u_1 - w_1 D_x u\|_2^2 \\ & + \frac{\tau}{2} \|u_2 - w_2 D_y u\|_2^2 \end{aligned}$$

where r_1 and r_2 are Lagrangian multipliers, $\tau > 0$. Model (12) is equivalent to the following model

$$\max_{r_1, r_2} \min_{u, u_1, u_2} L(u, u_1, u_2; r_1, r_2) \quad (13)$$

Model (13) can be split into the following subproblems:

$$\begin{aligned} u = & \arg \min_u L(u, u_1, u_2; r_1, r_2) \\ = & \arg \min_u \left\{ \frac{\lambda}{2} \|Bu - f\|_2^2 + \frac{\tau}{2} \|u_1 - w_1 D_x u + \frac{r_1}{\tau}\|_2^2 \right. \\ & \left. + \frac{\tau}{2} \|u_2 - w_2 D_y u + \frac{r_2}{\tau}\|_2^2 \right\} \end{aligned} \quad (14)$$

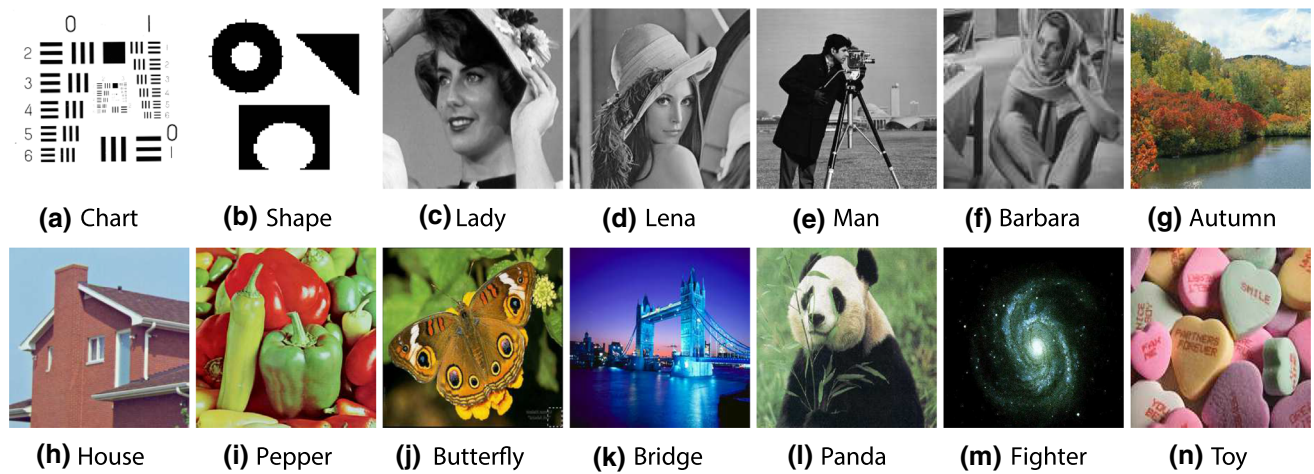


Fig. 1 Test images

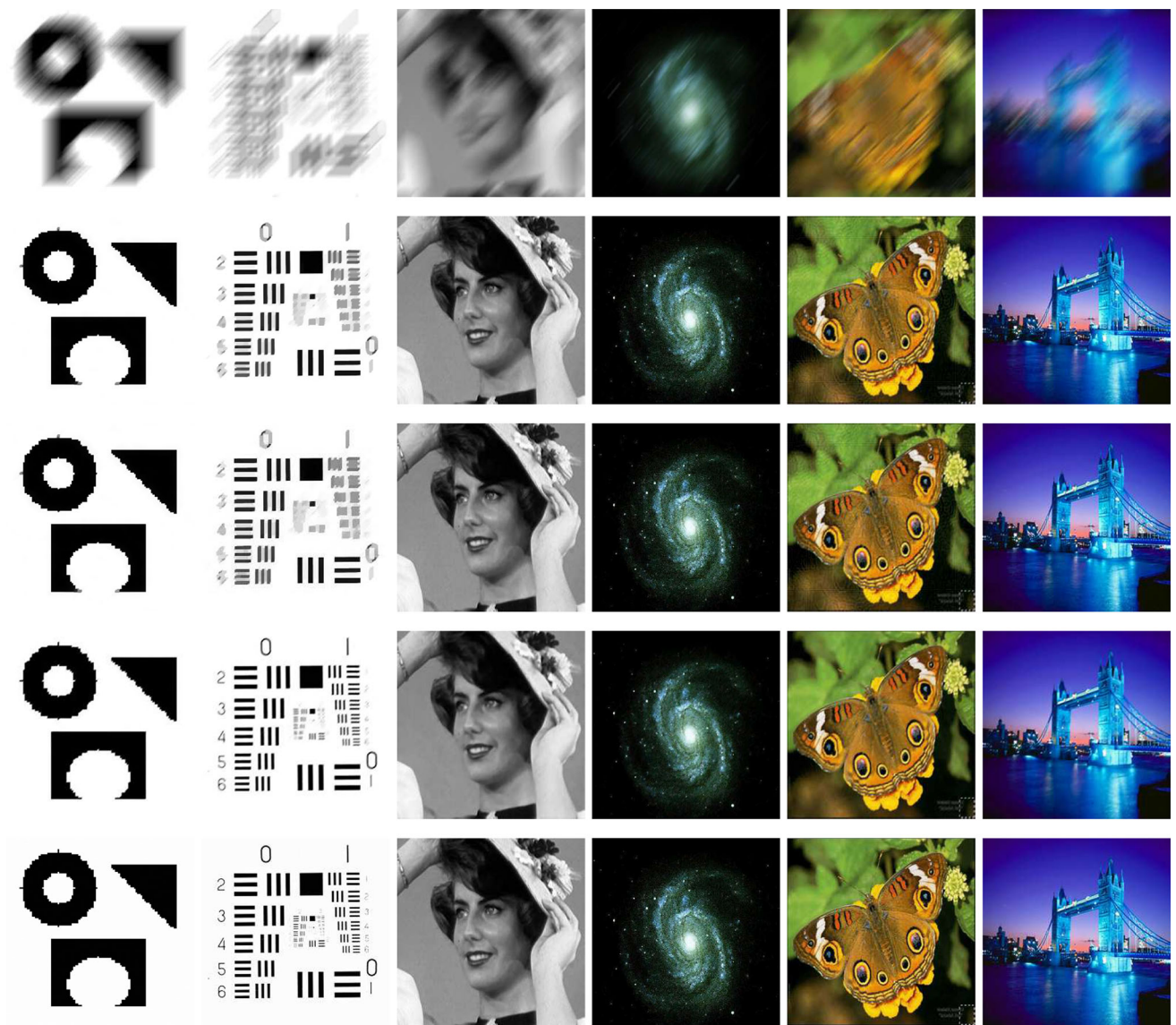


Fig. 2 Compared results of images with motion blur

$$u_1 = \arg \min_{b_1} L(u, u_1, u_2; r_1, r_2) \\ = \arg \min_{u_1} \{ \|u_1\|_1 + \frac{\tau}{2} \|u_1 - w_1 D_x u + \frac{r_1}{\tau}\|_2^2 \} \quad (15)$$

$$u_2 = \arg \min_{u_2} L(u, u_1, u_2; r_1, r_2) \\ = \arg \min_{u_2} \{ \|u_2\|_1 + \frac{\tau}{2} \|u_2 - w_2 D_y u + \frac{r_2}{\tau}\|_2^2 \} \quad (16)$$

Then the three subproblems can be solved successively. Minimizing (14) with respect to u ,

$$(\lambda B^* B + \tau w_1 D_x^* D_x + \tau w_2 D_y^* D_y) u \\ = \lambda B^* f + \tau w_1 D_x^* (u_1 + \frac{r_1}{\tau}) + \tau w_2 D_y^* (u_2 + \frac{r_2}{\tau})$$

where B^* is the adjoint operator of B . Let

$$T1 = \mathcal{F}(\lambda B^* B + \tau w_1 D_x^* D_x + \tau w_2 D_y^* D_y) \\ T2 = \mathcal{F}(\lambda B^* f + \tau w_1 D_x^* (u_1 + \frac{r_1}{\tau}) + \tau w_2 D_y^* (u_2 + \frac{r_2}{\tau}))$$

Therefore u can be computed by

$$u = \mathcal{F}^{-1} \left(\frac{T1}{T2} \right) \quad (17)$$

where $\mathcal{F}$ is the Fourier transform and $\mathcal{F}^{-1}$ is the inverse of $\mathcal{F}$. As for subproblems of u_1 and u_2 , they can be explicitly calculated by a generalized shrinkage [24] as follows

$$u_1 = \max(0, 1 - \frac{1}{\tau s}) s_1, u_2 = \max(0, 1 - \frac{1}{\tau s}) s_2. \quad (18)$$

where

$$s_1 = w_1 \cdot D_x u - \frac{r_1}{\tau}, s_2 = w_2 \cdot D_y u - \frac{r_2}{\tau}, s = \sqrt{(s_1^2 + s_2^2)}.$$

The problem (12) is iteratively updated by separately compute subproblems of (17)–(18) and multipliers r_1, r_2 , where the multipliers are updated by

$$r_1^{k+1} = r_1^k + \tau(u_1^k - w_1 \cdot D_x u^k) \\ r_2^{k+1} = r_2^k + \tau(u_2^k - w_2 \cdot D_y u^k) \quad (19)$$

k is the number of the k th iteration.

The stop of the iteration is up to the gap between two consecutive iterations. The stop criterion is written as

$$err := \frac{\|u^{k+1} - u^k\|_2}{\|u^k\|_2} \leq tol$$

where tol is the tolerance. It means that when the err is no more than tol after finite iterations then the algorithm will stop. Thus, a maximum of iterations is needed and denoted as $MaxIter$. The iterative steps are organized in algorithm 1.

3.3 Convergence analysis

In this subsection, the convergence of solutions generated from Algorithm 1 is analyzed.

Theorem 1 Suppose that $(u^*, u_1^*, u_2^*, r_1^*, r_2^*)$ is a saddle point of problem (13), then it is unique and the sequence $\{u^k, u_1^k, u_2^k, r_1^k, r_2^k\}$ generated by algorithm 1 converges to it.

Table 1 Comparison of SSIM and computational time (second) for deblurring

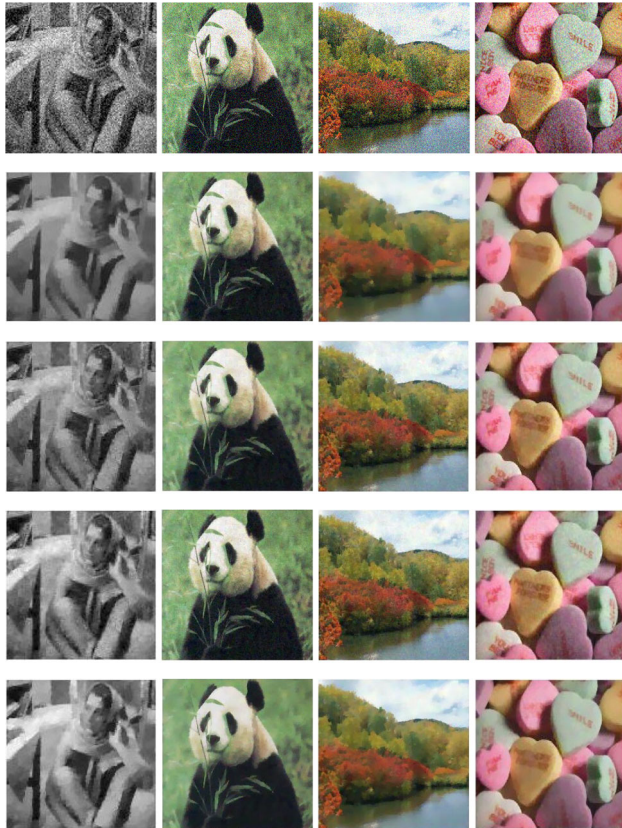
Image	SSIM				Time			
	L_1	$L_1-0.5L_2$	$HNSOTV_{OGS}$	WATV	L_1	$L_1-0.5L_2$	$HNSOTV_{OGS}$	WATV
Shape	0.9720	0.9652	0.9694	0.9752	0.8028	2.1755	2.0641	0.2662
Chart	0.8759	0.8543	0.9523	0.9956	1.0119	2.2953	2.1036	0.2790
Lady	0.9990	0.9990	0.9491	0.9999	0.2524	0.9842	7.0871	0.2836
Nebula	0.9675	0.9797	0.8723	0.9937	6.7182	8.6780	13.8390	0.6321
Butterfly	0.9387	0.9923	0.9739	0.9996	8.3456	13.9830	7.5863	0.3822
Bridge	0.9997	0.9984	0.9943	0.9999	0.3270	2.8599	4.9864	0.2883

Table 2 Compared SSIM and computational time (second) for denoising

Image	SSIM				Time			
	L_1	$L_1-0.5L_2$	$HNSOTV_{OGS}$	WATV	L_1	$L_1-0.5L_2$	$HNSOTV_{OGS}$	WATV
Barbara	0.7766	0.7757	0.7852	0.7993	0.1809	1.3847	0.6385	0.1385
Panda	0.8018	0.8248	0.7935	0.8342	2.6732	25.0924	8.8748	2.5117
Autumn	0.8723	0.8691	0.8733	0.8803	3.5657	23.6273	9.3613	3.2859
Toy	0.9028	0.9026	0.9007	0.9056	1.1211	7.3269	3.5585	1.0972

Algorithm 1 ADMM for image restoration

Initialize: $r_1^0, r_2^0, u_1^0, u_2^0$
Input: $\lambda, \alpha, \tau, MaxIter, tol, w_1, w_2$
Output: u
For $k = 1 : MaxIter$
 update u^{k+1} by Eq.(17)
 (u_1^{k+1}, u_2^{k+1}) by Eq.(18)
 $r_1^{k+1} \leftarrow r_1^k + \tau(u_1^k - w_1 \cdot D_x u^{k+1})$
 $r_2^{k+1} \leftarrow r_2^k + \tau(u_2^k - w_2 \cdot D_y u^{k+1})$
 if $err \leq tol$
 break
End

**Fig. 3** Compared results of image denoising

Proof The detail proof is referred to Ref. [43].

4 Experiments and results

In this section, the proposed WATV model is compared with L_1 and $L_1 - \alpha L_2$ -based models and HNSOTV-OGS model introduced in Sect. 2. They are applied to image deblurring, denoising and restoration (i.e., images contain simultaneously blur and noise) respectively. All the numerical experiments are performed via MATLAB (R2016a) on a windows7 (64bit) desktop computer with an Intel Core i5 3.30 GHz processor and 8.00 GB of RAM.

**Fig. 4** Compared results of images with motion blur and noise

The MATLAB codes for L_1 and $L_1 - \alpha L_2$ -based models are from authors of [28] (<https://sites.google.com/site/louyifei/>), HNSOTV-OGS is from the authors of [42]. Test images (shown in Fig. 1) of experiments include binary (a)–(b), gray (c)–(f) and color images (g)–(n).

In order to show the effectiveness and efficiency, qualities of restored results and computational time are provided. Qualities of images are evaluated by structural similarity index (SSIM), which is calculated via MATLAB function *ssim*. Larger SSIM means higher similarities of the results and the original images. The tolerance *tol* is defaulted to 10^{-4} , and the number of *MaxIter* is defaulted to 500 for all the numerical experiments.

Deblurring. When B in model (12) is a blur operator, the proposed WATV model is capable of deblurring. Parameters in WATV for removing motion blur are as follows: λ is default to 15, c is default to 10^{-4} and $\beta \in (0, 1)$. As seen in Fig. 2, the five rows from top to the bottom are blurred images, deblurring results of L_1 , $L_1 - \alpha L_2$, HNSOTV-OGS and WATV models, respectively. It is seen that the four models perform well in deblurring for gray and color images in vision. As for binary image Chart, HNSOTV-OGS and the proposed WATV model result in a more legible restoration. Besides visual effect, quantitative evaluation are also provided. The

Table 3 Compared SSIM and computational time (second) for restoration

Image	SSIM				Time			
	L_1	$L_1-0.5L_2$	$HNSOTV_{OGS}$	WATV	L_1	$L_1-0.5L_2$	$HNSOTV_{OGS}$	WATV
Lena	0.7272	0.7414	0.7195	0.7493	2.8621	4.4164	2.4216	2.0414
Man	0.7103	0.7193	0.7103	0.7247	2.1425	3.5867	2.4215	1.9053
House	0.9056	0.9115	0.9107	0.9197	4.2722	11.7619	10.7353	3.6414
Pepper	0.9445	0.9464	0.9476	0.9481	8.8413	17.9128	10.8533	7.4205

indicator SSIM and computational time are compared for deblurring in Table 1. It is seen that the WATV model obtains the highest SSIM and the least computational time, which shows the advantage over L_1 , $L_1-0.5L_2$ and HNSOTV-OGS models.

Denoising. In case of denoising, B in model (12) is an identical operator. In Fig. 3, one gray image (Barbara) and three color images (Panda, Autumn, Toy) are tested. In this application, λ is 6.5, 6, 5.5, 6 respectively, and c is default to 1 for WATV model. In Fig. 3, the five rows from top to the bottom are test images with white Gaussian noise of standard deviation 0.1, denoising results of L_1 , $L_1-0.5L_2$, HNSOTV-OGS and WATV models respectively. Besides visual effects, SSIM and computational time for quantitative evaluation are compared in Table 2. It can be seen that the WATV model results in the largest SSIM values and costs the least computational time.

Restoration. When images are simultaneously polluted by motion blur and white Gaussian noise, restored results are shown in Fig. 4. The five rows are degraded images, results of L_1 , $L_1-0.5L_2$, HNSOTV-OGS and WATV models respectively. In this application, λ is 35, 30, 35, 30 for WATV model, c belongs to the range of (0,1). In Table 3, WATV model results in the largest SSIM and costs the least computational time.

From experiments of image deblurring, denoising and restoration applications, the four compared models gain different effectiveness. The proposed WATV model exceeds other compared models in both SSIM and computational time. The L_1 performs better than $L_1-0.5L_2$ and in terms of computational time for deblurring, denoising and restoration. While HNSOTV-OGS model costs much more computational time. Therefore, based on the experiments and results above, the proposed WATV model performs better in effectiveness and efficiency.

5 Conclusion

In this paper, we propose a weighted anisotropic total variational model, which is applied to deblurring, denoising and restoration. In our model, a weighting function is designed

with respect to amplitudes of image gradients. The ADMM is explored for the proposed

model and the convergence is analyzed briefly. Compared experiments with L_1 , $L_1-0.5L_2$ and HNSOTV-OGS models are conducted, which show the efficiency and effectiveness of the proposed model in binary, gray and color images. In the following study, we will extend the proposed WATV model to other image problems, such as decomposition, bias correction and segmentation, etc.

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